

The game always ends because each of Sue's moves reduces the number of grey cells by at least one, and the initial number of grey cells is finite. Since the count of grey cells is a non-negative integer that strictly decreases with each of Sue's moves, the game cannot continue indefinitely.

To determine the number of lime cells when both players play optimally, note that the total number of cells, 2025×2025 , is odd. Linda can force the final move by ensuring the game ends on her turn. By strategically painting cells to maintain control over the parity of the remaining moves, Linda guarantees that the last grey cell is painted lime. Initially, there are an odd number of grey cells. Each pair of moves (Linda and Sue) reduces the number of grey cells by an even number, as Linda can choose to paint a grey cell (reducing the count by one, flipping parity) and Sue must reduce the count by at least one (resulting in a net reduction of even parity). This preserves the odd parity of the remaining grey cells until one grey cell remains. At this point, it is Linda's turn, and she paints the final cell lime. Thus, the number of lime cells equals half of the total cells rounded up, which is $\frac{2025^2+1}{2}$.

$$\boxed{\frac{2025^2 + 1}{2}}$$